Hybrid Triage System Architecture

How BioBERT NLP, Deterministic TEWS, and Bayesian Models Work Together — Including Voice and Twi Input

Quaigraine Samuel

Department of Mathematics
Kwame Nkrumah University of Science and Technology

May 19, 2026

Outline

The clinical problem

Hospital Chatbot for Patient Color-Coded Clinical Pathways

- Emergency departments in LMICs face **overcrowding** and the **Green Tag Burden** (non-urgent patients occupying acute resources).
- At KATH (Kumasi), mixed acuity and **under-triage** remain patient-safety risks.
- Many patients cannot self-report reliable vitals at home; they *can* describe symptoms in speech or SMS.

Solution: A **hybrid digital triage engine** that fuses language understanding with SATS mathematics *before* the patient reaches the ward.

Why a hybrid system?

No single model is safe alone:

Model	**Strength**	**Fatal flaw if used alone**
NLP (BioBERT)	Reads free text	Cannot verify vitals; slang risk
Deterministic (TEWS)	Objective, auditable	Fails when vitals missing
Bayesian	Handles uncertainty	Population priors may mislead

Design goal: Complementary layers with a **master controller** that prioritises patient safety (minimise under-triage).

Full equations, data contracts, and fusion spec: `biobert/mathematics/biobert-mathematics.pdf`.

How patients reach the system

Channel A: Text

- SMS, WhatsApp, web chat
- English or Twi (typed)

Channel B: Voice

- Microphone in app / USSD voice
- Twi or English spoken
- Speech $\rightarrow$ text $\rightarrow$ (translate) $\rightarrow$ NLP

Equity: Voice + local language support reduces literacy barriers; translation normalises input before BioBERT (trained primarily on English biomedical text).

End-to-end pipeline (conceptual)

1. **Capture** — patient speaks or types symptoms (and partial vitals if known).
2. **Pre-process** — ASR (voice), Twi$\rightarrow$English translation, text cleanup.
3. **Layer 1 (NLP)** — BioBERT $\rightarrow$ entities + SATS discriminators $\mathbf{D}$.
4. **Layer 2 (Deterministic)** — nurse/patient vitals $\rightarrow$ TEWS score T $\rightarrow$ base colour.
5. **Layer 3 (Bayesian)** — evidence $E \rightarrow P(C_k \mid E)$, infer missing risk.
6. **Fusion** — master rules $\rightarrow$ final pathway C (Red/Orange/Yellow/Green/Blue).
7. **Output** — colour code, queue instructions, alert to triage desk.

Data flow diagram

$$
\begin{array}{c}
\textbf{SMS/Chat} \rightarrow \text{normalise} \rightarrow \textbf{BioBERT} \rightarrow \textbf{D} \\
\downarrow \qquad \swarrow \qquad \searrow \\
\textbf{Voice} \rightarrow \text{ASR} \rightarrow \text{Translate(tw}\rightarrow\text{en)} \rightarrow \text{BioBERT} \\
\textbf{Vitals} \rightarrow \text{TEWS } (T) \qquad \textbf{Bayesian } (P) \\
\downarrow \qquad\qquad \downarrow \\
\textbf{Master controller} \rightarrow \text{Final colour } C
\end{array}
$$

All layers run in parallel where possible; fusion waits for available evidence (text always; vitals may arrive at nurse desk).

Voice pipeline overview

Step 1 — Automatic Speech Recognition (ASR):

- Convert audio waveform to transcript (Twi or English).
- Candidate service: **Google Cloud Speech-to-Text**.

Step 2 — Machine translation (if needed):

- If transcript language $\neq$ English, translate to English.
- Candidate service: **Google Cloud Translation API**.
- BioBERT NER/discriminators run on **English-normalised** text.

Step 3 — NLP triage: English text enters Layer 1 (BioBERT).

Google Cloud APIs: cost and access

Service	Role	Cost / access
Speech-to-Text	Audio $\rightarrow$ text	GCP account; free tier **limited** (not unlimited)
Translation API	tw $\rightarrow$ en	$\sim$**500k characters/month** free (NMT), then paid

Important

- APIs require a **Google Cloud project** with billing enabled (free tier still needs an account).
- Translation is **not permanently free** beyond the monthly character allowance.

Twi language: support caveat

Verify before deployment

Google's published Translation language list does **not clearly include Twi** (ISO 639-1: `tw`). Do *not* assume Twi→English works until tested on the live API.

Fallback strategies (FYP / pilot):

1. Bilingual UI: patient chooses **English** for full BioBERT path.
2. Twi ASR + **local glossary** (common symptom phrases → English).
3. Evaluate **alternate APIs** (e.g. Microsoft Translator language list for Akan/Twi).
4. SMS in Twi with human triage backup for unsupported paths.

Conceptual API sequence (implementation)

```
POST /speech/recognize  -> transcript, language_code
POST /translate         -> english_text (if not English)
POST /nlp/analyze       -> entities, discriminators D
POST /triage/compute    -> TEWS T, bayesian P, final_color C
```

Audit log: Store original Twi transcript + English translation + model outputs for clinical governance review.

Role of Layer 1 in the hybrid stack

Input: Normalised English symptom narrative (from SMS or voice pipeline).
Output: Structured clinical signal **D**:

- Named entities (symptoms, anatomy, severity)
- SATS **clinical discriminators** with confidence scores
- Optional: extracted numeric vitals mentioned in text ("pulse 125")

Does not: Replace TEWS arithmetic or issue a final diagnosis.

BioBERT pipeline (five steps)

1. **Tokenisation** (WordPiece) — sub-word units for rare medical terms.
2. **Embedding** — $E = E_{\text{word}} + E_{\text{position}} + E_{\text{segment}}$.
3. **Encoder** — L Transformer layers; contextual vectors per token.
4. **Attention** — high weight on clinically salient tokens (e.g. *crushing*, *chest*).
5. **Classification head** — map to SATS discriminator taxonomy.

BioBERT = BERT architecture + weights continually pre-trained on PubMed/PMC (Lee et al., 2020).

What BioBERT produces for fusion

Define discriminator vector:

$$\mathbf{D} = \{d_1, d_2, \ldots, d_m\}, \quad d_j \in [0, 1]$$

each $d_j =$ confidence that discriminator j is present (e.g. central chest pain, altered consciousness).

Hard safety rule (feeds master controller):

$$\text{if} \max_{j \in \text{Red/Orange}} d_j > \tau \Rightarrow \text{override low TEWS}$$

Threshold τ set by clinical governance (e.g. 0.85).

NLP when vitals are missing

Pre-hospital reality: Patients rarely own BP cuffs or pulse oximeters.

- Deterministic TEWS may be **incomplete** or zero.
- NLP still extracts urgency from language: *"chest is crushing, dizzy, cannot breathe"*.
- Bayesian layer uses NLP entities as evidence S for $P(D \mid S)$.

Complementarity: NLP active when algebra is blind; algebra anchors NLP when vitals exist.

Layer 1 failure modes (bounded by other layers)

- Colloquial Twi idiom mistranslated $\rightarrow$ mitigated by translation QA + glossary.
- Low-confidence NLP $\rightarrow$ Bayesian uses priors; nurse collects vitals.
- Hallucination risk $\rightarrow$ **fixed discriminator labels only** (no generative diagnosis text).

Role of Layer 2

Input: Objective vital signs v_k (mobility, RR, HR, temperature, AVPU, trauma).
Source:

- Patient-reported numbers in chat (if reliable), *or*
- **Triage nurse** measurement on arrival (primary clinical workflow).

Output: TEWS score T and base colour C_{TEWS}.

TEWS equation

Triage Early Warning Score (SATS)

$$T = \mathrm{TEWS} = \sum_{k=1}^{n} w_k \cdot f_k(v_k)$$

- f_k: piecewise scoring function per vital (0–3 points)
- w_k: protocol weights
- Fully **deterministic**: same inputs $\Rightarrow$ same T (auditable, no randomness)

Example (heart rate): $f_1(v_1) = 3$ if $v_1 \geq 130$; 0 if $51 \leq v_1 \leq 100$.

Colour mapping from TEWS

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T > 7 \\ \text{Orange} & 5 \le T \le 6 \\ \text{Yellow} & 3 \le T \le 4 \\ \text{Green} & T \le 2 \end{cases}$$

Limitation: If any required v_k is missing, T may be undefined or under-estimated $\rightarrow$ Layer 3 $+$ NLP overrides required.

Dynamic re-calculation

- Initial chatbot session may be **text-only** (T unknown).
- Nurse enters vitals at desk $\rightarrow$ system **recalculates** T instantly.
- Colour can **upgrade** (e.g. Green $\rightarrow$ Orange) if vitals worsen.

Integration point: Layer 2 receives structured vitals from the same patient record ID as Layer 1 text session.

Role of Layer 3

Input: Evidence E = NLP entities $\cup$ partial vitals $\cup$ demographics.
Output:

- Posterior $P(D \mid S)$ for latent severity / disease D
- $P(C_k \mid E)$ over colour classes when evidence incomplete
- **Clinical override** signal to master controller

Bayes' theorem in triage

$$P(D \mid S) = \frac{P(S \mid D)\,P(D)}{P(S)}$$

- $P(D)$: **prior** (disease prevalence at hospital)
- $P(S \mid D)$: **likelihood** (symptoms given disease)
- $P(D \mid S)$: **posterior** — updated urgency belief

Use case: Normal vitals but text suggests sepsis/MI $\rightarrow$ high posterior $\Rightarrow$ upgrade pathway.

Missing vitals and incomplete evidence

When temperature or BP is unknown:

- Deterministic layer cannot complete T.
- Bayesian network **infers** probable acuity from remaining E.
- Master rule: if evidence incomplete,

$$C = \arg\max_k P(C_k \mid E)$$

Safety bias: Priors and discriminators tuned to prefer **over-triage** over under-triage where ambiguous.

Scenario: septic shock override

Patient text: *"I am confused, freezing cold, have not urinated today."*

- TEWS from sparse vitals: Green
- $P(\text{Septic shock} \mid S) \approx 0.92$
- **Action:** Bayesian + discriminator rules $\Rightarrow$ Red override

Under-triage requires **both** deterministic and probabilistic layers to fail; NLP discriminators add a third safety net.

Master controller: priority rules

Let $C \in \{\text{Red}, \text{Orange}, \text{Yellow}, \text{Green}, \text{Blue}\}$.

1. If **D** contains Red/Orange discriminator with confidence $> \tau \Rightarrow C = \text{Red or Orange}$.
2. Else if $T \geq 7 \Rightarrow C = \text{Red}$.
3. Else if $T \in [5, 6] \Rightarrow C = \text{Orange}$.
4. Else if $T \in [3, 4] \Rightarrow C = \text{Yellow}$.
5. Else if evidence incomplete $\Rightarrow C = \arg\max_k P(C_k \mid E)$.
6. Else $C = \text{Green}$.

Conflict resolution

NLP **D**	TEWS T	**Winner**
High (chest pain)	Low ($T = 4$)	NLP discriminator
Low	High ($T \geq 7$)	TEWS
Ambiguous	Incomplete	Bayesian
Low	Low	Green (with disclaimer)

Principle: When in doubt, route to the **more urgent** colour (SATS philosophy).

Why the ensemble is safer

- **NLP** catches language-only emergencies (MI, stroke phrases).
- **TEWS** anchors decisions to physiology when data exist.
- **Bayesian** fills gaps when data are partial.
- **Voice + translate** extends access without bypassing the same three layers.

Research: multimodal triage (text + vitals + probability) outperforms unimodal models in admission prediction and resource forecasting (Gligorijević et al., 2018; Stewart et al., 2023).

Outputs to the patient and hospital

- **Patient:** colour explanation, expected wait guidance, when to seek immediate help.
- **Triage desk:** C, T, top discriminators, transcript (Twi + English), confidence scores.
- **Specialist alert:** Orange/Red cardiac or trauma flags to dashboard.

Disclaimer: Decision support only — final triage by qualified staff.

Scenario A: English SMS (chest pain) — Part 1

Input: *"Chest is crushing. Dizzy. Pulse fast, about 125. Breathing heavy, 26 per minute. No thermometer."*

Layer 1 (BioBERT):

- Entities: chest pain, dizziness, tachycardia, tachypnoea
- **D**: Central chest pain (high confidence)

Layer 2 (TEWS): HR=125 $\rightarrow$ 2 pts; RR=26 $\rightarrow$ 2 pts; $\Rightarrow$ $T = 4$ (Yellow alone).

Scenario A: English SMS — Part 2 (fusion)

Layer 3 (Bayesian): Missing BP/temperature $\Rightarrow$ high $P(\text{cardiac event} \mid E)$.

Master decision

Discriminator *central chest pain* overrides $T = 4$.
Final: Orange pathway + cardiology alert.

Same logic as comprehensive example in literature-review slides (Kumasi SMS case).

Scenario B: Twi voice input

1. **Voice (Twi):** Patient speaks symptom phrase.
2. **ASR:** Twi transcript (e.g. fever, chest pain, breathlessness colloquialism).
3. **Translate:** English: *"I have fever, my chest hurts, I am short of breath."*
4. **BioBERT:** Flags fever, chest pain, respiratory distress; high attention on *chest*, *breath*.
5. **Fusion:** If discriminators for cardio-respiratory distress $\Rightarrow$ urgent colour even before nurse vitals.

Original Twi example from project nlp-short deck; translation step made explicit here.

Scenario C: TEWS Green but NLP urgent

- Nurse measures vitals: all normal $\Rightarrow T = 0$, $C_{\text{TEWS}} =$ Green.
- Patient chat: *"Worst headache of my life, stiff neck, vomiting."*
- BioBERT: possible CNS red-flag discriminator.
- Bayesian: high $P(\text{meningitis/SAH} \mid S)$.

Final: Override Green $\Rightarrow$ Red/Orange.
Demonstrates why TEWS alone is insufficient in pre-hospital chatbot use.

Modular backend architecture

- **Ingestion service:** SMS, web, voice endpoints
- **Language service:** ASR + translation adapters (swappable if Twi unsupported)
- **NLP service:** BioBERT inference (GPU optional for prototype)
- **Triage engine:** TEWS calculator + Bayesian network + fusion rules
- **Persistence:** session store, audit logs, pathway definitions (SATS tables)

Each module exposes JSON; fusion is a single deterministic function of module outputs.

Ethics, limitations, and next steps

Limitations:

- Translation errors for Twi; domain shift vs. PubMed training
- Self-reported symptoms; no physical examination
- Google API cost and language support must be validated in pilot

Next steps:

- Pilot at KATH with nurse-in-the-loop validation
- Fine-tune BioBERT on local de-identified chat logs
- Evaluate Twi fallbacks; add Ga, Hausa per main-presentation roadmap

References

- Lee, J., et al. (2020). BioBERT. *Bioinformatics*, 36(4), 1234–1240.
- Devlin, J., et al. (2018). BERT. arXiv:1810.04805.
- Gligorijević, D., et al. (2018). Deep attention model for ED triage. arXiv:1804.03240.
- Stewart, J., et al. (2023). NLP at ED triage: narrative review. *PLoS ONE*, 18(12):e0279953.
- Google Cloud. Speech-to-Text & Translation API documentation (pricing, language support).
- SATS manual; Rominski et al. KATH validation (project papers).
- Project decks: `biobert/mathematics/`, `latex/nlp-short/`, `latex/literature-review/`.